The Journal of Open Source Software

popclass: a python package for classifying microlensing events

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DOI: 10.xxxxxx/draft

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Editor: Ivelina Momcheva

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- @skterry
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Submitted: 07 October 2024
Published: unpublished

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Summary

popclass is a python package that provides a flexible, probabilistic framework for classifying the lens of a gravitational microlensing event. Gravitational microlensing occurs when a massive foreground object (e.g., a star, white dwarf or a black hole) passes in front of and lenses the light from a distant background star. This causes an apparent brightening, and shift in position, of the background source. In most cases, characteristics of the microlensing signal do not contain enough information to definitively identify the lens type. However, different lens types can lie in different but overlapping regions of the characteristics of the microlensing signal. For example, black holes tend to be more massive than stars and therefore cause microlensing signals that are longer. Current Galactic simulations allow the prediction of where different lens types lie in the observational space and can therefore be leveraged to classify events (e.g., Lam et al., 2020).

popclass allows a user to match characteristics of a microlensing signal to a simulation of the Galaxy to calculate lens type probabilities for an event (see Figure 1). Constraints on any microlensing signal characteristics and any Galactic model can be used. popclass comes with an interface to ArviZ (Kumar et al., 2019) and pymultinest (Buchner, 2016) for microlensing signal constraints, pre-loaded Galactic models, plotting functionality, and classification uncertainty quantification methods. The probabilistic framework for popclass was developed in Perkins et al. (2024), used in Fardeen et al. (2024) and has been applied to classifying events in Kaczmarek et al. (2024).

Statement of need

The advent of the Vera C. Rubin Observatory (Ivezić et al., 2019) and the *Nancy Grace Roman Space Telescope* (Spergel et al., 2015) will provide tens-of-thousands of microlensing events per year (e.g., Abrams et al., 2023; Penny et al., 2019). To maximize the science output of this event stream it is critical to identify events that have a high probability of being caused by a lens type such as a black hole (Lam et al., 2022; Sahu et al., 2022), and then to allocate expensive follow-up observations such as space-based astrometry (e.g., Sahu et al., 2022) or ground-based adaptive optics imaging (e.g., Terry et al., 2022) to confirm their nature.

Current microlensing software packages such as DarkLensCode (Howil et al., 2024) or PyLiMASS (Bachelet et al., 2024) estimate lens mass-distance constraints using microlensing event light curve and additional auxiliary information (e.g., source proper motions, distances, color, or finite source effects). Using auxiliary information makes these current methods powerful but

limits them to only be effective for events with the available auxiliary data. Moreover, no current software tools explicitly predict lens type, and they always assume a fixed Galactic model. `popclass` fills the need for a flexible microlensing classification software package that can be broadly applied to classify all events from the Vera C. Rubin Observatory and *Nancy Grace Roman Space Telescope* and can be used with any Galactic model in the form of a simulation.

Method

`popclass` relies on the Bayesian classification framework in (Perkins et al., 2024). Consider the data from a single microlensing event $\boldsymbol{d}$, and a model of the Galaxy $\mathcal{G}$. `popclass` calculates the probability that the lens of the event belongs to each lens class, class_L, where $\text{class}_L \in \text{classes}$ and, for example, $\text{classes} = \{\text{Star, Neutron Star, White Dwarf, Black Hole}\}$. Namely, `popclass` calculates

$$p(\text{class}_L|\boldsymbol{d}, \mathcal{G}) \text{ for } \text{class}_L \in \text{classes}.$$

Using Bayes' theorem,

$$p(\text{class}_L|\boldsymbol{d}, \mathcal{G}) = \frac{p(\text{class}_L|\mathcal{G})p(\boldsymbol{d}|\text{class}_L, \mathcal{G})}{p(\boldsymbol{d}|\mathcal{G})}.$$

Assuming that the set of considered lens classes is complete, $p(\boldsymbol{d}|\mathcal{G})$ is a normalization factor such that all lens class probabilities sum to unity. Using importance sampling (e.g., Hogg et al., 2010) with S independent posterior samples $\theta_c \sim p(\theta|\boldsymbol{d})$ drawn under some prior, $\pi(\theta)$, obtained from fitting some set of microlensing signal parameters, θ,

$$p(\text{class}_L|\boldsymbol{d}, \mathcal{G}) = \frac{p(\text{class}_L|\mathcal{G})}{p(\boldsymbol{d}|\mathcal{G})} \times \frac{1}{S}\sum_{c=0}^{S} \frac{p(\theta_c|\text{class}_L, \mathcal{G})}{\pi(\theta_c)}$$

.

This allows the use of previously calculated posterior samples to perform lens classification for a single event in the context of a Galactic model. The term $p(\theta_c|\text{class}_L, \mathcal{G})$ is calculated using kernel density estimation in `popclass` over θ with a simulated catalog of microlensing events from $\mathcal{G}$. $p(\text{class}_L|\mathcal{G})$ is the prior probability that an event belongs to each class before any data is seen, which is just set by relative number of expected events predicted by the Galactic model $\mathcal{G}$.

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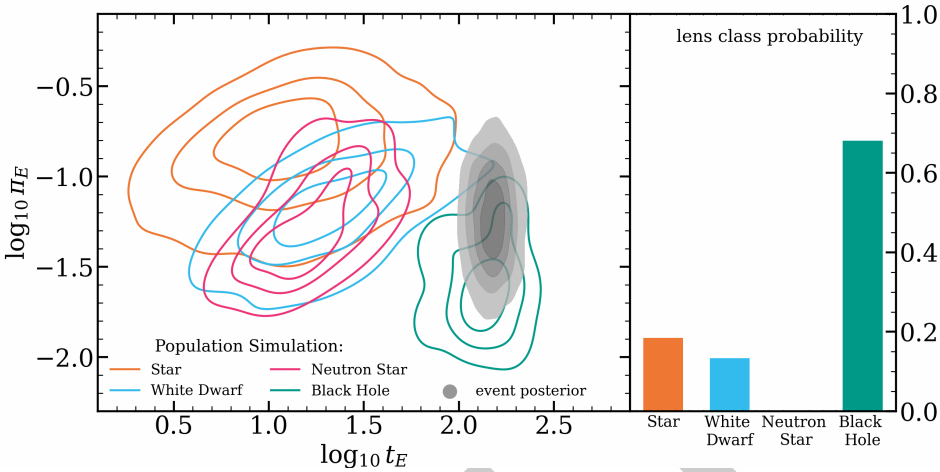

Figure 1: Left: posterior distribution of an event in log10(timescale)-log10(parallax) space, overlaid on 'star', 'white dwarf', 'neutron star' and 'black hole' contours. Right: bars showing probabilities of that event belonging to each of the lens populations.

Acknowledgements

popclass depends on numpy (Harris et al., 2020), scipy (Virtanen et al., 2020), asdf (Greenfield et al., 2015), matplotlib (Hunter, 2007), and scikit-learn (Buitinck et al., 2013). This work was performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory (LLNL) under Contract DE-AC52-07NA27344. The document number is LLNL-JRNL-870290 and the code number is LLNL-CODE-2000456. The theoretical foundation of this work was established under support from Lawrence Livermore National Laboratory's Laboratory Directed Research and Development Program under project 22-ERD-037. The software implementation for this project was funded under the LLNL Space Science Institute's Institutional Scientific Capability Portfolio funds in partnership with LLNL's Academic Engagement Office. ZK acknowledges support from the 2024 LLNL Data Science Summer Institute and is a Fellow of the International Max Planck Research School for Astronomy and Cosmic Physics at the University of Heidelberg (IMPRS-HD).

References

Abrams, N. S., Hundertmark, M. P. G., Khakpash, S., Street, R. A., Jones, R. L., Lu, J. R., Bachelet, E., Tsapras, Y., Moniez, M., Blaineauu, T., Di Stefano, R., Makler, M., Varela, A., & Rabus, M. (2023). Microlensing Discovery and Characterization Efficiency in the Vera C. Rubin Legacy Survey of Space and Time. *arXiv e-Prints*, arXiv:2309.15310. https://doi.org/10.48550/arXiv.2309.15310

Bachelet, E., Hundertmark, M., & Calchi Novati, S. (2024). Estimating Microlensing Parameters from Observables and Stellar Isochrones with pyLIMASS. *168*(1), 24. https://doi.org/10.3847/1538-3881/ad4862

Buchner, J. (2016). *PyMultiNest: Python interface for MultiNest*. Astrophysics Source Code Library, record ascl:1606.005.

Buitinck, L., Louppe, G., Blondel, M., Pedregosa, F., Mueller, A., Grisel, O., Niculae, V., Prettenhofer, P., Gramfort, A., Grobler, J., Layton, R., VanderPlas, J., Joly, A., Holt, B., & Varoquaux, G. (2013). API design for machine learning software: Experiences from the scikit-learn project. *ECML PKDD Workshop: Languages for Data Mining and Machine*

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Learning, 108–122.

Fardeen, J., McGill, P., Perkins, S. E., Dawson, W. A., Abrams, N. S., Lu, J. R., Ho, M.-F., & Bird, S. (2024). Astrometric Microlensing by Primordial Black Holes with the Roman Space Telescope. *965*(2), 138. https://doi.org/10.3847/1538-4357/ad3243

Greenfield, P., Droettboom, M., & Bray, E. (2015). ASDF: A new data format for astronomy. *Astronomy and Computing*, *12*, 240–251. https://doi.org/10.1016/j.ascom.2015.06.004

Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk, M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

Hogg, D. W., Myers, A. D., & Bovy, J. (2010). Inferring the Eccentricity Distribution. *725*(2), 2166–2175. https://doi.org/10.1088/0004-637X/725/2/2166

Howil, K., Wyrzykowski, Ł., Kruszyńska, K., Zieliński, P., Bachelet, E., Gromadzki, M., Mikołajczyk, P. J., Jabłońska, M., Kaczmarek, Z., Mróz, P., Ihanec, N., Ratajczak, M., Pylypenko, U., Rybicki, K., Sweeney, D., Hodgkin, S. T., Larma, M., Carrasco, J. M., Burgaz, U., … Morrell, M. (2024). Uncovering the Invisible: A Study of Gaia18ajz, a Candidate Black Hole Revealed by Microlensing. *arXiv e-Prints*, arXiv:2403.09006. https://doi.org/10.48550/arXiv.2403.09006

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55

Ivezić, Ž., Kahn, S. M., Tyson, J. A., Abel, B., Acosta, E., Allsman, R., Alonso, D., AlSayyad, Y., Anderson, S. F., Andrew, J., Angel, J. R. P., Angeli, G. Z., Ansari, R., Antilogus, P., Araujo, C., Armstrong, R., Arndt, K. T., Astier, P., Aubourg, É., … Zhan, H. (2019). LSST: From Science Drivers to Reference Design and Anticipated Data Products. *873*(2), 111. https://doi.org/10.3847/1538-4357/ab042c

Kaczmarek, Z., McGill, P., Perkins, S., Dawson, W., Abrams, N., Ho, M., Huston, M., & Lu, J. (2024). *On Finding Black Holes in Photometric Microlensing Surveys*.

Kumar, R., Carroll, C., Hartikainen, A., & Martin, O. (2019). ArviZ a unified library for exploratory analysis of bayesian models in python. *Journal of Open Source Software*, *4*(33), 1143. https://doi.org/10.21105/joss.01143

Lam, C. Y., Lu, J. R., Hosek, Jr., Matthew W., Dawson, W. A., & Golovich, N. R. (2020). PopSyCLE: A New Population Synthesis Code for Compact Object Microlensing Events. *889*(1), 31. https://doi.org/10.3847/1538-4357/ab5fd3

Lam, C. Y., Lu, J. R., Udalski, A., Bond, I., Bennett, D. P., Skowron, J., Mróz, P., Poleski, R., Sumi, T., Szymański, M. K., Kozłowski, S., Pietrukowicz, P., Soszyński, I., Ulaczyk, K., Wyrzykowski, Ł., Miyazaki, S., Suzuki, D., Koshimoto, N., Rattenbury, N. J., … Terry, S. K. (2022). An Isolated Mass-gap Black Hole or Neutron Star Detected with Astrometric Microlensing. *933*(1), L23. https://doi.org/10.3847/2041-8213/ac7442

Penny, M. T., Gaudi, B. S., Kerins, E., Rattenbury, N. J., Mao, S., Robin, A. C., & Calchi Novati, S. (2019). Predictions of the WFIRST Microlensing Survey. I. Bound Planet Detection Rates. *241*(1), 3. https://doi.org/10.3847/1538-4365/aafb69

Perkins, S. E., McGill, P., Dawson, W., Abrams, N. S., Lam, C. Y., Ho, M.-F., Lu, J. R., Bird, S., Pruett, K., Golovich, N., & Chapline, G. (2024). Disentangling the Black Hole Mass Spectrum with Photometric Microlensing Surveys. *961*(2), 179. https://doi.org/10.3847/1538-4357/ad09bf

Sahu, K. C., Anderson, J., Casertano, S., Bond, H. E., Udalski, A., Dominik, M., Calamida, A., Bellini, A., Brown, T. M., Rejkuba, M., Bajaj, V., Kains, N., Ferguson, H. C., Fryer, C. L.,

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Yock, P., Mróz, P., Kozłowski, S., Pietrukowicz, P., Poleski, R., … RoboNet Collaboration. (2022). An Isolated Stellar-mass Black Hole Detected through Astrometric Microlensing. *933*(1), 83. https://doi.org/10.3847/1538-4357/ac739e

Spergel, D., Gehrels, N., Baltay, C., Bennett, D., Breckinridge, J., Donahue, M., Dressler, A., Gaudi, B. S., Greene, T., Guyon, O., Hirata, C., Kalirai, J., Kasdin, N. J., Macintosh, B., Moos, W., Perlmutter, S., Postman, M., Rauscher, B., Rhodes, J., … Zhao, F. (2015). Wide-Field InfrarRed Survey Telescope-Astrophysics Focused Telescope Assets WFIRST-AFTA 2015 Report. *arXiv e-Prints*, arXiv:1503.03757. https://doi.org/10.48550/arXiv.1503.03757

Terry, S. K., Bennett, D. P., Bhattacharya, A., Koshimoto, N., Beaulieu, J.-P., Blackman, J. W., Bond, I. A., Cole, A. A., Lu, J. R., Marquette, J. B., Ranc, C., Rektsini, N., & Vandorou, A. (2022). Adaptive Optics Imaging Can Break the Central Caustic Cusp Approach Degeneracy in High-magnification Microlensing Events. *164*(5), 217. https://doi.org/10.3847/1538-3881/ac9518

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … SciPy 1. 0 Contributors. (2020). SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*, 261–272. https://doi.org/10.1038/s41592-019-0686-2

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